COMPLETION ALONG A CLOSED SUBSET OF A MODIFICATION SMOOTH OVER THE COMPLEMENT IS RIG-SMOOTH

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a double etale affine covering reduction, a locality argument for rig-smooth morphisms of formal algebraic spaces, and an affine algebra conclusion via the I-adic completion criterion, all inside the delivered span

Editorial note. The label lemma-smooth-gives-rig-smooth names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The proof opens with the author's sentence 'We urge the reader to skip the details.' and then nevertheless delivers the complete reduction and affine argument; per the quality criterion this reading suggestion is not an omission, and the verification is delivered verbatim.

Allowed standard prerequisites (editorial).

- The modification setting: f : X' to X with U' to U smooth over the complement of T, and cartesianity of the diagrams formed by completing along inverse images of T
- The covering definition of locally Noetherian formal algebraic spaces, and rig-smoothness being local on the source and target for morphisms of formal algebraic spaces
- The affine algebra criterion identifying rig-smoothness of completed finite type ring maps

Source reference. Original definition 0AIM: sources/stacks-chow/tags.txt, source lines 13270–13270. Prerequisite pointer invoked at proof line(s) 6151; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AQ1: sources/stacks-chow/tags.txt, source lines 13494–13494. Prerequisite pointer invoked at proof line(s) 6177; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAK: sources/stacks-restricted/pinned-original-source.tex, source lines 943–943. Prerequisite pointer invoked at proof line(s) 6132; it names a standard fact of the same chapter that the counted argument uses, and

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every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCQ: sources/stacks-restricted/pinned-original-source.tex, source lines 4642–4642. Prerequisite pointer invoked at proof line(s) 6152; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03FX: sources/stacks-chow/tags.txt, source lines 4508–4508. Prerequisite pointer invoked at proof line(s) 6137; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In the situation above. If X is locally Noetherian, f is locally of finite type, and $U' \to U$ is smooth, then $f_{/T}$ is rig-smooth.*

Proof. The strategy of the proof is this: reduce to the case where X and X' are affine, translate the affine case into algebra, and finally apply Lemma 0GAK. We urge the reader to skip the details.

Choose a surjective étale morphism $W \to X$ with $W = \coprod W_i$ a disjoint union of affine schemes, see Properties of Spaces, Lemma 03FX. For each i choose a surjective étale morphism $W_i' \to W_i \times_X X'$ where $W_i' = \coprod W_{ij}'$ is a disjoint union of affines. In particular $\coprod W_{ij}' \to X'$ is surjective and étale. Denote $f_{ij} : W_{ij} \to W_i$ the given morphism. Denote $T_i \subset W_i$ and $T_{ij}' \subset W_{ij}$ the inverse images of T. Since taking the completion along the inverse image of T produces cartesian diagrams (see above) we have $(W_i)_{/T_i} = W_i \times_X X_{/T}$ and similarly $(W_{ij}')_{/T_{ij}'} = W_{ij}' \times_{X'} X'_{/T'}$. Moreover, recall that $(W_i)_{/T_i}$ and $(W_{ij}')_{/T_{ij}'}$ are affine formal algebraic spaces. Hence $\{(W_{ij}')_{/T_{ij}'} \to X'_{/T'}\}$ is a covering as in Formal Spaces, Definition 0AIM. By Lemma 0GCQ we see that it suffices to prove that

$$(W_{ij}')_{/T_{ij}'} \longrightarrow (W_i)_{/T_i}$$

is rig-smooth. Observe that $W_{ij}' \to W_i$ is locally of finite type and induces a smooth morphism $W_{ij}' \backslash T_{ij}' \to W_i \backslash T_i$ (as this is true for f and these properties of morphisms are étale local on the source and target). Observe that W_i is locally Noetherian (as X is locally Noetherian and this property is étale local on the algebraic space). Hence it suffices to prove the lemma when X and X' are affine schemes.

Assume $X = \mathrm{Spec}(A)$ and $X' = \mathrm{Spec}(A')$ are affine schemes. Since X is Noetherian, we see that A is Noetherian. The morphism f is given by a ring map $A \to A'$ of finite type. Let $I \subset A$ be an ideal cutting out T. Then IA' cuts out T'. Also $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is smooth over $\mathrm{Spec}(A) \setminus T$. Let $A^\wedge$ and $(A')^\wedge$ be the I-adic completions. We have $X_{/T} = \mathrm{Spf}(A^\wedge)$ and $X'_{/T'} = \mathrm{Spf}((A')^\wedge)$, see proof of Formal Spaces, Lemma 0AQ1. By Lemma 0GAK we see that $(A')^\wedge$ is rig-smooth over $(A.I)$ which in turn means that $A^\wedge \to (A')^\wedge$ is rig-smooth which finally implies that $X'_{/T'} \to X_{/T}$ is rig smooth by Lemma 0GCQ. □